Project3

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1 Project Three: DTT Power exhaust modelling

In this project, a DTT 2D power exhaust modelling case is provided. The deuterium and another impurity are considered. The total power entering the Scrape-Off Layer is 30MW. Please answer the following question according to the corespoing plots from the modelling results.

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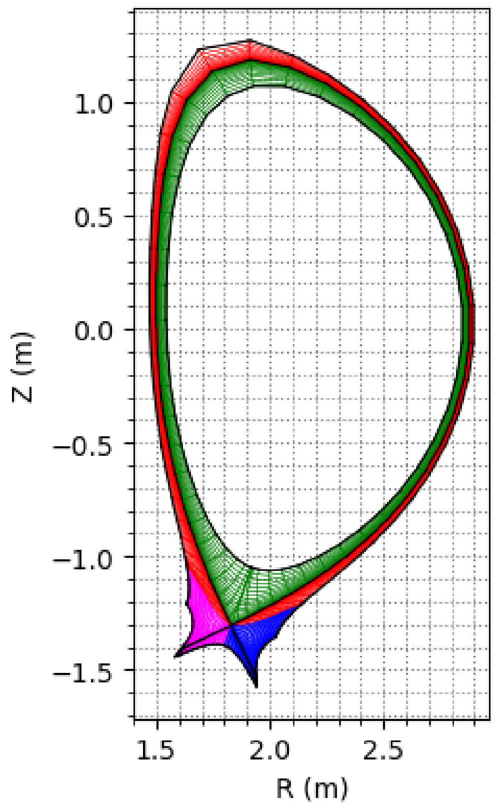
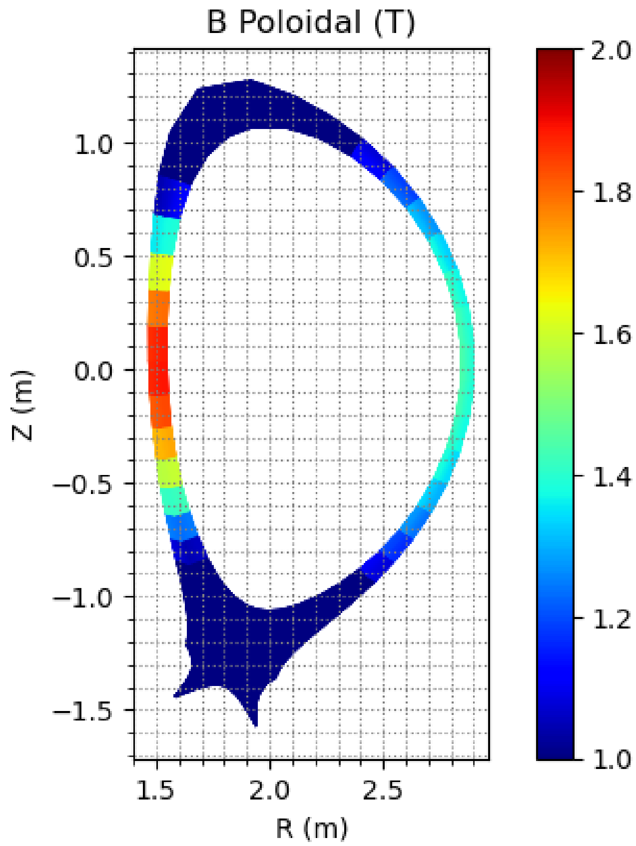
[30]:
```python
# In this part, we import necessary enviromental variables and functions,
# which are used in this hand-on session.
import matplotlib.pyplot as plt
import numpy as np
import math
import sys
sys.path.append('/home/jovyan/pex2024')
import sptools.read as spr
import sptools.plot as spp
```

[36]:
```python
AUG_b2fgmtry_file = './Data/b2fgmtry'
gmtry = spr.read_b2file(AUG_b2fgmtry_file)
spp.plot_geometry(gmtry)
B_pol = gmtry['bb'][:, :, 0].reshape(98, 38)
spp.plot_2dfield(gmtry,B_pol,figname='B Poloidal (T)',scale='linear')
B_tor = np.abs(gmtry['bb'][:, :, 2].reshape(98, 38))
spp.plot_2dfield(gmtry,B_tor,figname='B Toroidal (T)',scale='linear')

#Question 1-3
#Geometric data
#Major radius
RR = 2.2
#minor radius
aa = 0.7

#Q.1
```

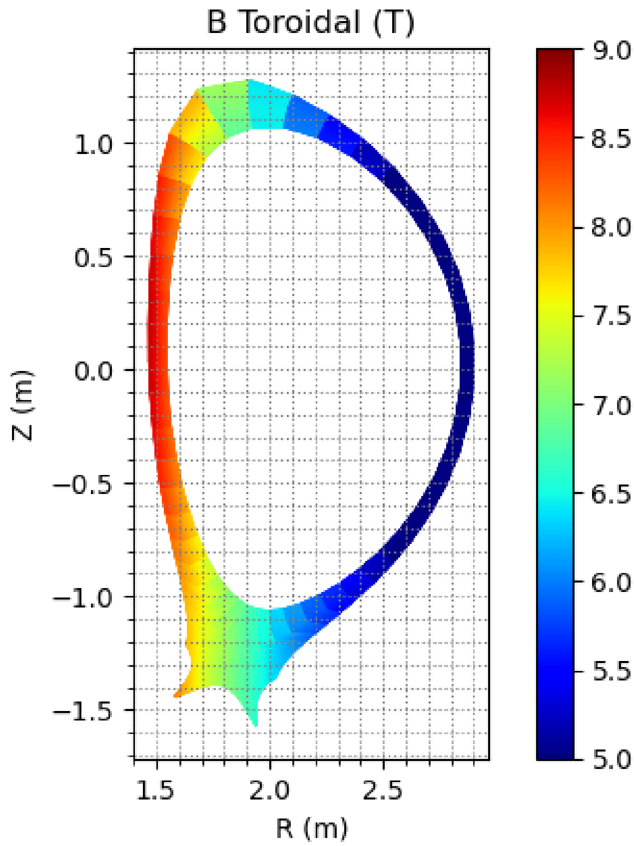
```python
#Since the particles, before hitting the wall, will go through the all tokamak,
↪I consider that they will face an average poloidal and toroidal magnetic field
B_pol_mean=np.mean(B_pol)
B_pol_OMP = 1.4
B_tor_mean=np.mean(B_tor)
#safety factor
q = aa*B_tor_mean/(RR*B_pol_mean)
#connection length
LL = np.pi*RR*q

print('Average poloidal magnetic field -->',f'{B_pol_mean:.2f}','T')
print('Average toroidal magnetic field -->',f'{B_tor_mean:.2f}','T')
print('Safety factor -->',f'{q:.2f}')
print('Connection length -->',f'{LL:.2f}','m')

#Q.2
Te = 150*1.602e-19
v_th_i = np.sqrt(2*Te/3.34e-27)
larmor_rho = v_th_i*3.34e-27/(1.6e-19*B_pol_mean)
larmor_rho_OMP = v_th_i*3.34e-27/(1.6e-19*B_pol_OMP)
#SOL width
lambda_q = 2*aa*larmor_rho/RR
lambda_q_OMP =  2*aa*larmor_rho_OMP/RR

print('Average SOL width -->', f'{lambda_q*1e3:.2f}','mm')
print('SOL width at OMP -->', f'{lambda_q_OMP*1e3:.2f}','mm')

#Q.3
ne = 6e19
lambda_mfp_OMP = 1e16*150**2/ne
knudsen_num = lambda_mfp_OMP/(LL)
print('the mean free path is',f'{lambda_mfp_OMP:.2f}','m')
print('the knudsen number is', f'{knudsen_num:.2f}')
```



1.0
0.5
0.0
−0.5
−1.0
−1.5
Z (m)
1.5
2.0
2.5
R (m)



B Poloidal (T)
1.0
0.5
0.0
−0.5
−1.0
−1.5
Z (m)
1.5
2.0
2.5
R (m)
2.0
1.8
1.6
1.4
1.2
1.0




```
Average poloidal magnetic field --> 0.68 T
Average toroidal magnetic field --> 6.87 T
Safety factor --> 3.21
Connection length --> 22.17 m
Average SOL width --> 2.34 mm
SOL width at OMP --> 1.14 mm
the mean free path is 3.75 m
the knudsen number is 0.17
```

Question 1: Based on the above plots, including the computational mesh and the poloidal and toroidal magnetic field distribution, please estimate the safety factor q and the connection length, and explain their physical meanings.

1. Looking at the graphs of the magnetic fields, we can see that the toroidal field has a larger range of variation with respect to the poloidal one and, as we expected, $B\Phi$ is greater since it is the main component of the magnetic confinement, with a net proportionality $B\Phi \propto 1/R$. The safety factor (q) represents the number of turns of the toroidal field lines per turn of the poloidal field line and is calculated as:

$$q = \frac{aB\Phi}{RB\theta} \tag{1}$$

where a is the minor radius and R the major radius, respectively:

$$a = 0.7\ m$$
$$R = 2.2\ m$$

In general, $q > 1$ is needed to maintain a stable core plasma and in some configurations, like this one, the edge plasma can reach $q > 3$. The particles in the edge plasma overall, follow the magnetic field lines but their toroidal and poloidal components continuosly change across the tokamak; therefore, I compute the average value of $B\Phi$ and $B\theta$ that the particles feel.

$$B\Phi_{mean} = 6.87\ T$$
$$B\theta_{mean} = 0.68\ T$$

Using the previous data, I obtain a safety factor $q = 3.2$, which is a feasible value since in the edge plasma it is essential to maintain the plasma stability, reducing interactions with the first wall. Nevertheless, in a fixed geometry, the safety factor cannot be increased arbitrarily, in fact, reducing $B\theta$ affects the stability and $B\Phi$ has technical limits, the superconductors cannot exceed the critical values (I_c and B_c) and it must be take into account electromechanical stresses due to $J \times B$ forces.

The connection length is strictly connected with the safety factor. In a divertor configuration, following a magnetic field line, it represents the distance between the upstream region, where ideally the particles enter the SOL, and the outer/inner divertor target. It is a useful quantity to build models of the SOL, like the 2-Point Model.

The general formula to compute the connection length is:

$$2L = 2\pi R q \tag{2}$$

where $2L$ is the target-to-target distance along a magnetic field line, so the connection length is half of this quantity.

$$L = 22.2\ m$$

Question 2: If the Temperature is assumed to be 150 eV, please estimate the power flux width λ_q and explain the reason.

2. The power flux width, also called SOL power width is the width of the channel through which the exhausted power from the core that enters the SOL travels and hits the divertor targets. It also represents the decay constant of the power flux parallel to the magnetic field at the Outer Mid Plane.

$$q_{//} = q_0\ e^{-x/\lambda_q} \tag{3}$$

The electron density and the electron temperature have the same exponential behaviour, so in general we can also define λ_n and λ_T. In this case, it is assumed that they have the same value. The following formula (4) estimates the width based on a 2-D model of the SOL, assuming a circular plasma cross-section and evaluating the SOL width as the distance reached by the ions bounced back from the X-point to the OMP, due to the pressure imbalance generated by ΔB and curvature drift.

$$\lambda_q = \frac{2a\rho_p}{R} \tag{4}$$

where ρ_p is the Larmor radius of the deuterium ions:

$$\rho_p = \frac{v_{th,i}\ m_i}{eB\theta} \tag{5}$$

$m_i = 3.34 \cdot 10^-27\ kg$, deuterium ion mass

$v_{th,i} = \sqrt{\frac{2T_i}{m_i}} = 119.9\ km/s$, thermal ion velocity

The SOL width at the OMP, with $B\theta = 1.4\ T$, is $\lambda_{q,OMP} = 1.14\ mm$, in accordance with what we expected. Furthermore, considering the average value of the poloidal magnetic field $B\theta = 0.68\ T$, corresponds to $\lambda_q = 2.34\ mm$, this is because of the dependence of the Larmor radius on $B\theta$ ($\rho_p \propto \frac{1}{B\theta}$). Hence, the power flux width, near the X-point and the divertor, will be larger since $B\theta$ is lower.

Question 3: The modeling results are based on the fluid plasma transport equations. If the upstream density is 6e19 m^{-3}, explain why the fluid method is reasonable for describing plasma transport in this case.

3. The type of model to choose depends on the collisional mean-free-path, which is the distance covered by a particle between two collisions. Computing this quantity in the upstream region with $n_e = 6 \cdot 10^{-19}\ m^{-3}$ and $T_e = 100\ eV$, it is obtained:

$$\lambda_{mfp} = \frac{10^{16}\ Te^2}{n_e} = 3.75\ m \tag{6}$$

The fluid model approach can be confirmed using a dimensionless quantity, the Knudsen number, defined as:

$$Kn = \frac{\lambda_{mfp}}{2L} \tag{7}$$

The denominator of (7) is the target-to-target distance and represents the characteristic length of the 1-D model of the SOL that you may use considering a fluid method. If the length scale of the SOL system is in the same range of the mean free path, i.e. $kn = 1$, the plasma cannot be treated as a continuum. In general, for Knudsen numbers above 0.1, the fluid model could fail and a kinetic approach must be taken into account.

In this case, $kn = 0.085$, so a fluid method is reasonable but the mean free path strictly depends on the region in which it is computed and the Knudsen number depends on the spatial scale of the system analyzed. For example, if the analysis is focused on the divertor region, a kinetic model could be needed; in fact, the plasma interaction with the wall and the increasing presence of different particle species in a small volume could lead to Knudsen number larger than one.

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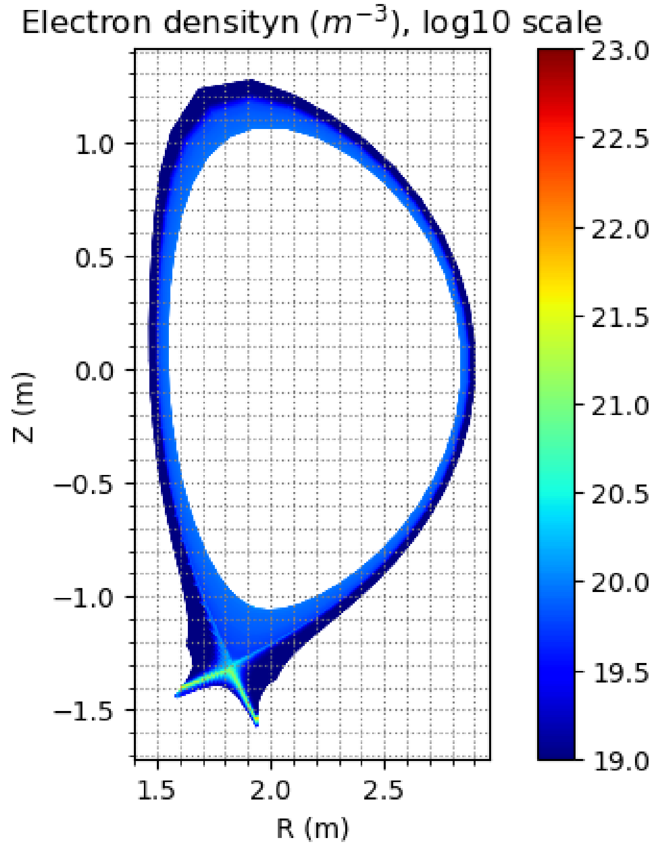
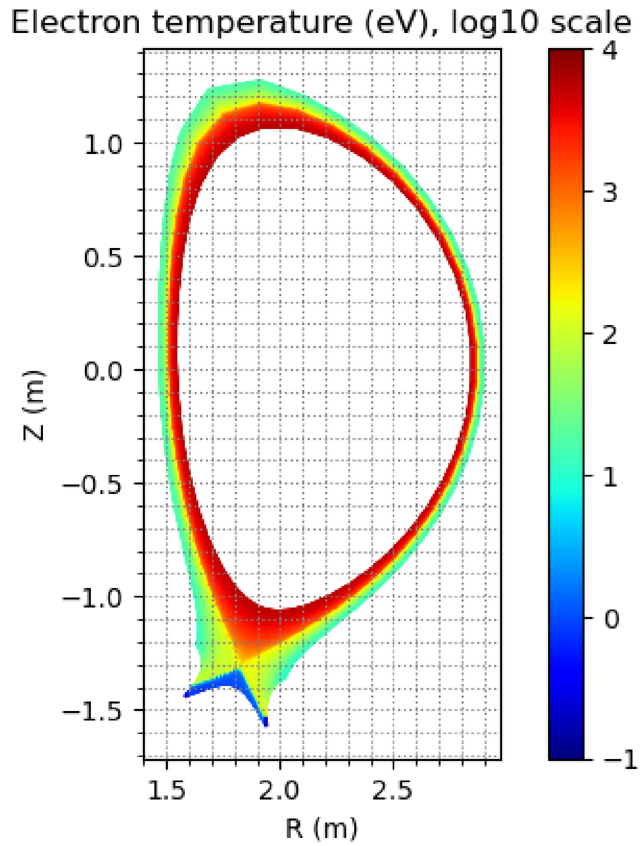
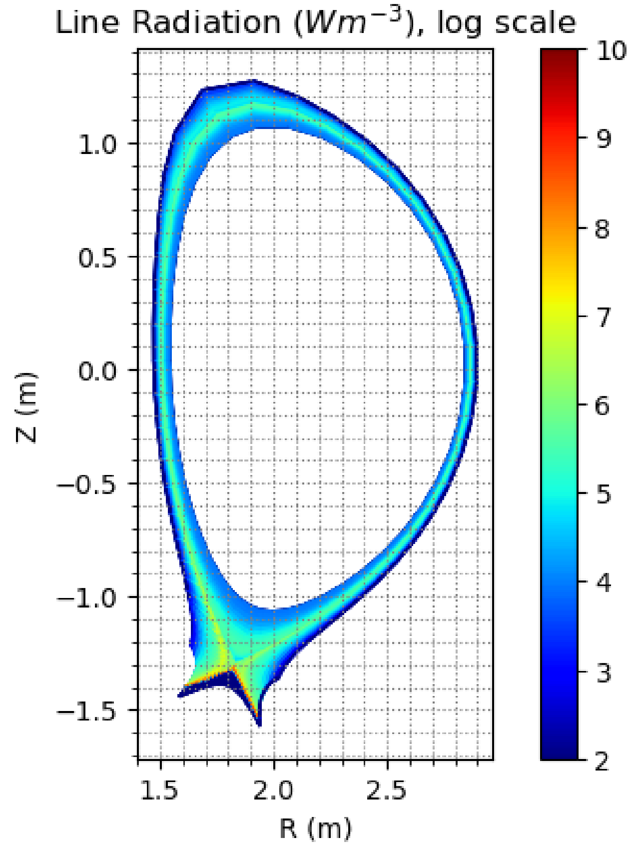
[45]:
```
gmtry = spr.read_b2file('./Data/b2fgmtry')
state = spr.read_b2file('./Data/b2fplasmf')
spp.plot_2dfield(gmtry,state['ne'],figname='Electron densityn ($m^{-3}$), log10␣
↪scale')
```

```
spp.plot_2dfield(gmtry,state['te']/1.602e-19,figname='Electron temperature (eV),␣
↪log10 scale')

rqrad = np.sum(state['rqrad'],axis=2)
vol = state['vol']
rqrad = np.clip(rqrad, 1e-5, None)
rad = np.abs(rqrad)/vol
spp.plot_2dfield(gmtry,rad,figname='Line Radiation ($W m^{-3}$), log scale')

#Q.4
#total line radiation
tot_lrad = np.sum(rad*vol)
print('Total line radiation -->',f'{tot_lrad*1e-6:.2f}','MW')

#ionization energy
eps=13.5*1.602e-19
gamma = 9
#electron density at the target
ne_t = 5e19
#electron temperature at the target
Te_t = 10*1.602e-19
#electron pressure
pe_t = ne_t*Te_t
#power flux at the wall
q_wall = pe_t*((2/3.34e-27)**0.5)*(gamma*(Te_t**0.5)+eps/(Te_t**0.5))
print('Total power flux to the wall -->',f'{q_wall*1e-6:.2f}','MW/m^2')
```


Electron densityn (m^{-3}), log10 scale
Z (m)
R (m)
1.0
0.5
0.0
−0.5
−1.0
−1.5
1.5
2.0
2.5
23.0
22.5
22.0
21.5
21.0
20.5
20.0
19.5
19.0


Electron temperature (eV), log10 scale
Z (m)
1.0
0.5
0.0
−0.5
−1.0
−1.5
1.5
2.0
2.5
R (m)
4
3
2
1
0
−1




```
Total line radiation --> 10.56 MW
Total power flux to the wall --> 25.68 MW/m^2
```

Question 4: The above plots show the electron density, electron temperature, and line radiation distributions. 1) Explain what line radiation is. 2) Estimate the total line radiation. 3) Estimate the peak value of the total power flux (including plasma contribution and surface recombination contribution) and explain the reason. 4) Determine whether the power exhaust problem is resolved in this case, and explain your reason.

4. One of the possible ways to reduce the heat flux to the divertor targets is to increase the radiative part, because radiation is an isotropic phenomenon and due to this, the heat can be dissipated in a larger volume.

Line radiation is one of the radiative phenomena that may happen in the edge plasma and it occurs when orbital electrons, after being excited to a higher orbital, return to the normal state in the lowest possible orbital. It is typically triggered by an encounter with a free electron.

The energy needed to move electrons between orbitals is quantized ($\Delta E = h\nu$) and an atom, returning to its ground state, releases this energy in the form of radiation. Therefore, the atomic spectrum is discretized, hence the name 'line radiation'.

This phenomenon is used to dissipate more heat, injecting purposely some impurities, noble gases in particular, inside the edge plasma. Each element has its particular electrons distribution therefore its specific spectrum and the important parameter to quantify how much power a particular element can dissipate is called radiating power (L_z).

The total line radiation power is computed by making the integral over the total domain; since the geometry is discretized, I compute the total sum of the line radiation density times each volume, obtaining $tot_{rad} = 10.56\ MW$. It represents one-third of the total power entering from the core, which clearly confirms the importance of line radiation and, in general, how much the radiative mechanisms can help for the Power EXhaust problem. We must be careful because they also represent drawbacks when heavy atoms like tungsten sputters from the wall penetrate into the core plasma and with the same mechanism cooling down the plasma. This is because impurity transport is another of the main topics of fusion reactors.

Analyzing the distribution of the electron density and the electron temperature, it can be said that the DTT works in the sheath-limited regime since, along a flux tube (parallel direction), there is not a big density gradient between the upstream and the divertor region and the temperature is quite constant through the SOL.

This SOL regime is characterized by high input power, the direct consequence is a high temperature, but since the heat conductivity coefficient has a temperature proportionality $\chi_e \propto T^{5/2}$, so there is the possibility to accommodate large heat fluxes with moderate or negligible temperature gradients. Consequently, the temperature target is determined by the sheath edge, in particular by the sheath heat transmission factor $\gamma = 9$.

$$q_{wall,plasma} = \gamma\ T_e\ n_{se}\ c_s \tag{8}$$

where n_{se} is the electron density at the sheath edge entrance and c_s is the sound velocity (from the Bohm criterion).

The total heat flux takes into account also the heat flux deposited by the electron-ion pair recombination at the wall, obtaining a hydrogen atom. The ionization energy spent to originally split them in the plasma is released onto the wall.

$$q_{wall,recomb} = n_{se}\ c_s\ \epsilon \tag{9}$$

$\epsilon = 13.5\ eV$, ionization energy.

This two contribution have been summed up to evaluate the total power flux, considering the peak value just after the separatrix in the outer divertor target.

$$q_{wall,peak} = p_e \sqrt{\frac{2}{m_i}}\ (\gamma\sqrt{T_e}\ + \frac{\epsilon}{\sqrt{T_e}}) \tag{10}$$

This formula is obtained from the 2-Point model where the electron pressure $p_e = n_e\ T_e$ has to be conserved along a flux tube, so that this value can be considered a constant; this assumption seems to be not too far from the reality having regard to the previous consideration on the sheath-limited regime.

As peak value, I select from the graphs $n_e = 5 \cdot 10^{19}$ and $T_e = 10\ eV$, leading to:

$$q_{wall,peak} = 25.68\ MW/m^2$$

Since the engineering benchmark with actual technologies of the divertor is around $10MW/m^2$ in steady state operation, it can be said that the power exhaust problem is not solved.

However, the sheath-limited regime is not the best working condition option, instead, the detachment could be a better solution. It occurs at low temperature, leading to momentum and energy losses due to different phenomena such as volumetric recombination and ion-neutral friction. The injection of impurities could be an option to decrease the temperature through radiation, triggering detachment.